

SKYNETIX

Enterprise Multi-Branch Network Design and Simulation

Cisco Packet Tracer

Network Scope	Four branch locations
Address Space	200.168.30.0/24
Routing	Static routing
Core Technologies	IPv4, VLSM, LAN/WAN, ICMP

1. Project Overview

SkyNetix is a simulated enterprise network connecting four branch locations in Malaysia: Kuala Lumpur, Penang, Johor Bahru, and Kota Kinabalu. The network was implemented in Cisco Packet Tracer to demonstrate practical LAN and WAN design, structured IPv4 addressing, VLSM subnetting, static routing, and end-to-end connectivity.

2. Network Objectives

- Design a complete LAN and WAN topology for four geographically separated branches.
- Allocate IPv4 addresses efficiently using Variable Length Subnet Masking (VLSM).
- Configure branch routers, departmental switches, servers, and end devices.
- Establish point-to-point WAN connectivity between branch routers.
- Configure static routes for communication between remote networks.
- Verify network operation using intra-branch and inter-branch ICMP tests.

3. Overall Network Topology

The network consists of four branch LANs interconnected through WAN links. Within each branch, departmental devices connect through access switches to the branch router. The branch routers form a four-site WAN ring using point-to-point serial links.

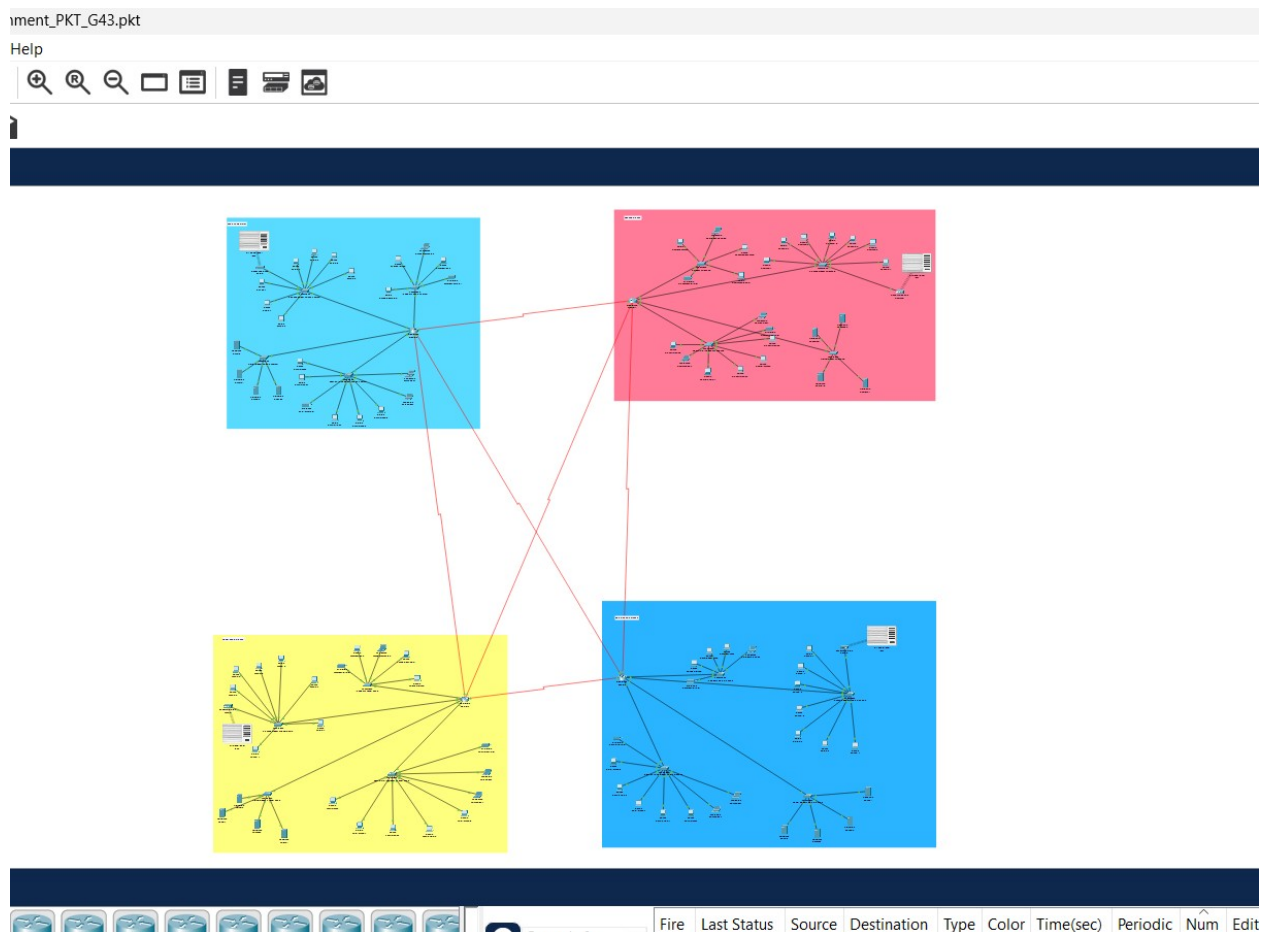


Figure 1. Complete four-branch topology in Cisco Packet Tracer.

4. LAN and WAN Architecture

4.1 Departmental LANs

Each branch uses a star-based departmental LAN structure. PCs, laptops, printers, servers, and other endpoints connect to local switches. The branch router provides the default gateway for each departmental subnet and forwards traffic to other networks.

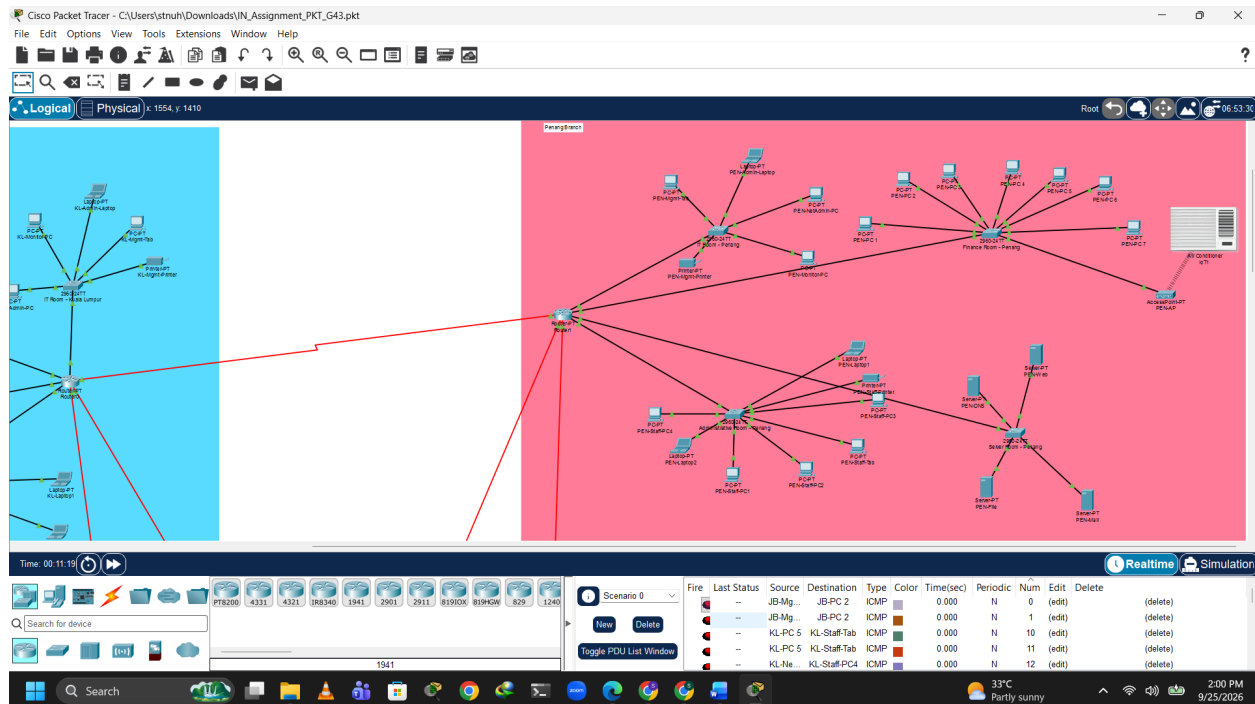


Figure 2. Penang branch topology showing departmental LANs, router, switches, servers, and end devices.

4.2 WAN Connectivity

The four routers are connected through four point-to-point links: Kuala Lumpur-Penang, Penang-Johor Bahru, Johor Bahru-Kota Kinabalu, and Kota Kinabalu-Kuala Lumpur. Each WAN segment uses a /30 subnet, which provides two usable IPv4 addresses.

5. IP Addressing and VLSM

The addressing plan uses the 200.168.30.0/24 network. Each branch is assigned a /26 block. Within the branch blocks, larger departments use /28 networks and smaller IT and Server segments use /29 networks. The remaining address space is used for /30 WAN links.

Branch	Address Block	Function
Kuala Lumpur	200.168.30.0/26	Headquarters
Penang	200.168.30.64/26	Regional branch
Johor Bahru	200.168.30.128/26	Regional branch
Kota Kinabalu	200.168.30.192/26	Regional branch

Prefix	Total Addresses	Usable Hosts	Use
/28	16	14	Finance and Administration
/29	8	6	IT and Server
/30	4	2	WAN point-to-point links

6. WAN Addressing Plan

Link	Subnet	Router Interface Addresses	Broadcast
KL - Penang	200.168.30.240/30	.241 / .242	200.168.30.243
Penang - Johor Bahru	200.168.30.244/30	.245 / .246	200.168.30.247
Johor Bahru - Kota Kinabalu	200.168.30.248/30	.249 / .250	200.168.30.251
Kota Kinabalu - KL	200.168.30.252/30	.253 / .254	200.168.30.255

7. Routing and Device Configuration

Each router interface is configured with an address from the relevant LAN or WAN subnet. The lowest usable address in each LAN subnet is used as the default gateway. Static routes are configured on the routers to provide reachability to remote branch networks.

Static route syntax: ip route [destination network] [subnet mask] [next-hop address]

Departmental switches provide Layer 2 connectivity. End devices are configured with static IPv4 addresses, subnet masks, default gateways, and DNS settings where required. Server devices represent services such as web, file, mail, print, and DNS.

8. Connectivity Testing

The completed topology was tested using ICMP in Cisco Packet Tracer. Tests were performed between devices in the same branch and between devices located in different branches. Successful tests demonstrate correct end-device addressing, default-gateway configuration, router interface addressing, WAN connectivity, and static routing.

During validation, an unsuccessful Johor Bahru ICMP attempt was identified and corrected before successful connectivity was confirmed. This reflects the normal troubleshooting process used when validating network configurations.

Validation Area	Method	Outcome
Intra-branch connectivity	ICMP ping between local devices	Successful
Inter-branch connectivity	ICMP ping across WAN links	Successful
Johor Bahru troubleshooting	ICMP retest after correcting configuration	Successful after correction

9. Technical Skills Demonstrated

Cisco Packet Tracer | IPv4 Addressing | VLSM | Subnetting | Static Routing | LAN/WAN Design | Star Topology | Point-to-Point Links | Default Gateways | ICMP Testing | Network Troubleshooting | Technical Documentation

10. Potential Enhancements

- Introduce VLANs for logical department segmentation.
- Apply access control lists (ACLs) to control inter-network traffic.
- Use a dynamic routing protocol to improve scalability.
- Add redundant network paths and switching infrastructure.
- Introduce network security monitoring and additional service validation.

11. Conclusion

The SkyNetix network demonstrates a complete four-site enterprise network simulation. The design combines departmental LANs, VLSM-based IPv4 allocation, point-to-point WAN links, static routing, server connectivity, and practical ICMP validation. The implementation provides a clear demonstration of foundational network design, configuration, and troubleshooting skills.